

Re-cap of Ethical Risk

- Behavioural datasets contain quasi-identifiers
- Combining attributes increases re-identification risk
- Amazon review dataset shows high uniqueness levels
- Behavioural patterns create digital fingerprints

Up to 14.9% of users become uniquely identifiable when multiple quasi-identifiers are combined.

Mitigation Strategy Overview

- Three mitigation techniques were implemented:
 - Data generalisation
 - k-anonymity suppression
 - Differential privacy noise injection
- **Goal:**
Reduce re-identification risk while preserving data utility for analytics

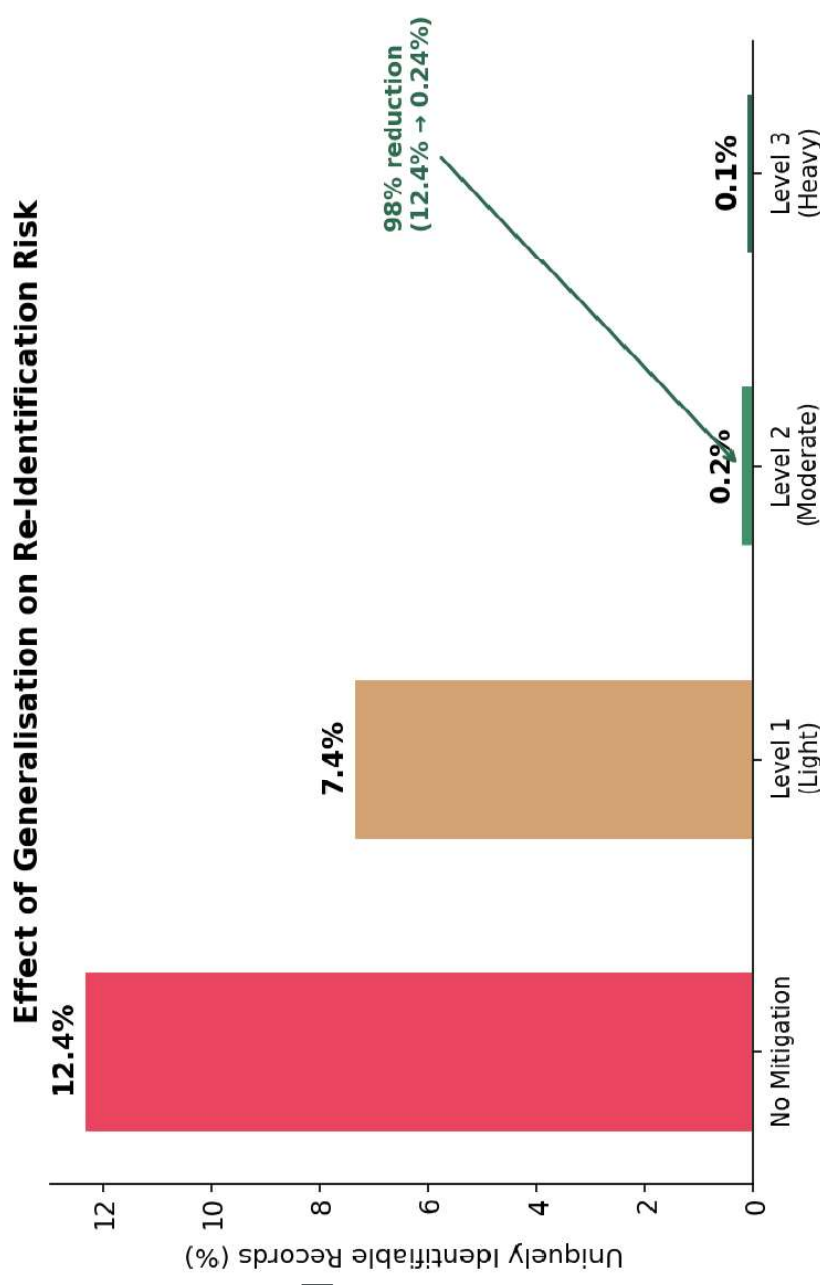
Mitigation Strategy 1

Data generalization

- Sensitive attributes were **generalised** to **reduce uniqueness**
- Examples:
 - Country → Regional groups
 - Hour of review → Time of day categories
 - Review length → Text length bins

Effect

- Reduced uniqueness from **12.4% to 0.24%**.



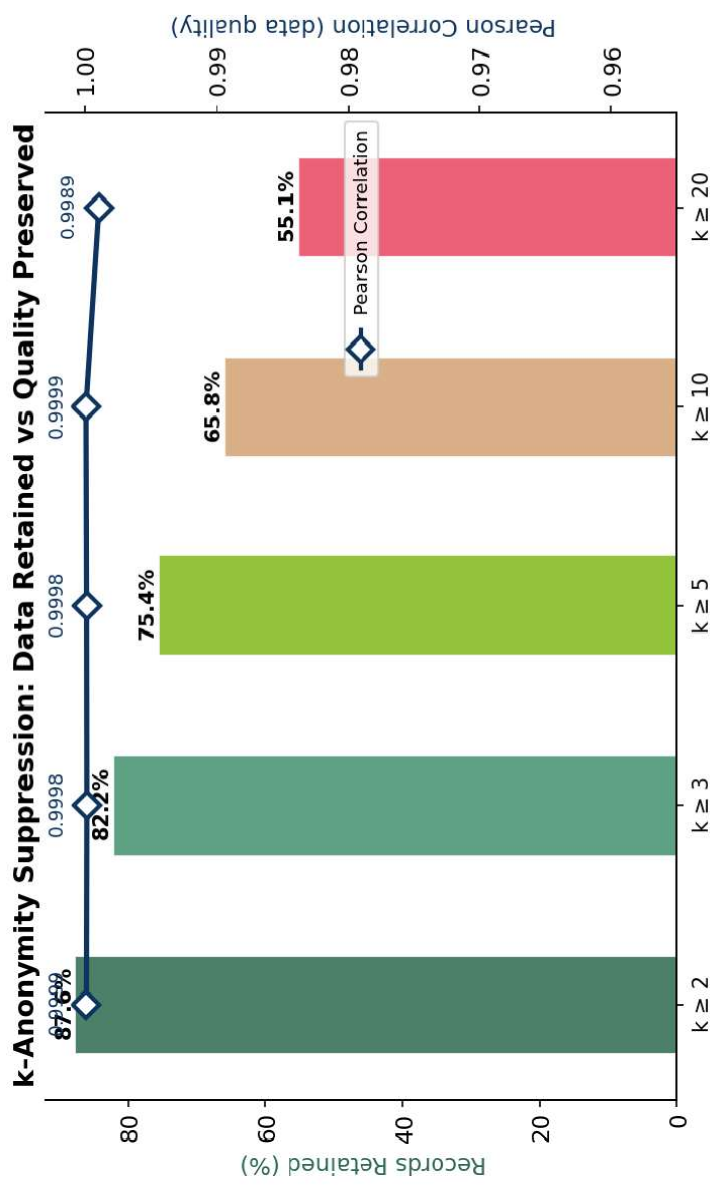
Mitigation Strategy 2

k-Anonymity Suppression:

- Dataset was evaluated using k-anonymity thresholds
- Records must belong to a group of $k \geq 5$ individuals
- Highly unique records were suppressed

Result

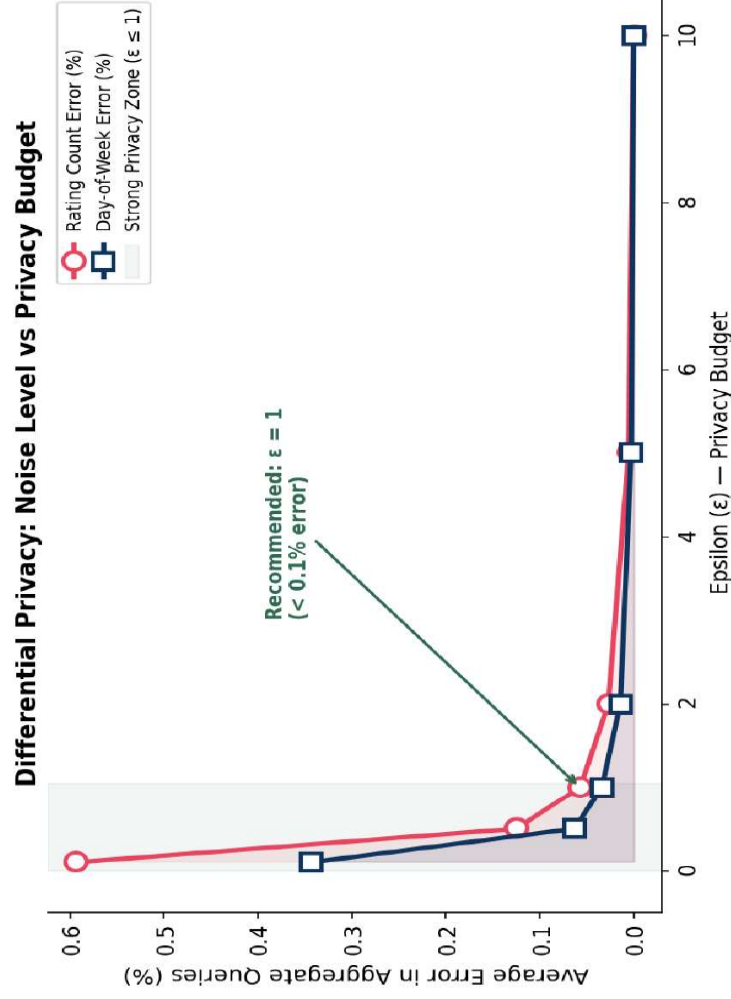
- 75.4% of dataset retained
- Analytical distribution preserved
- Pearson correlation = **0.9998**



Mitigation Strategy 3

Differential Privacy and Synthetic Data:

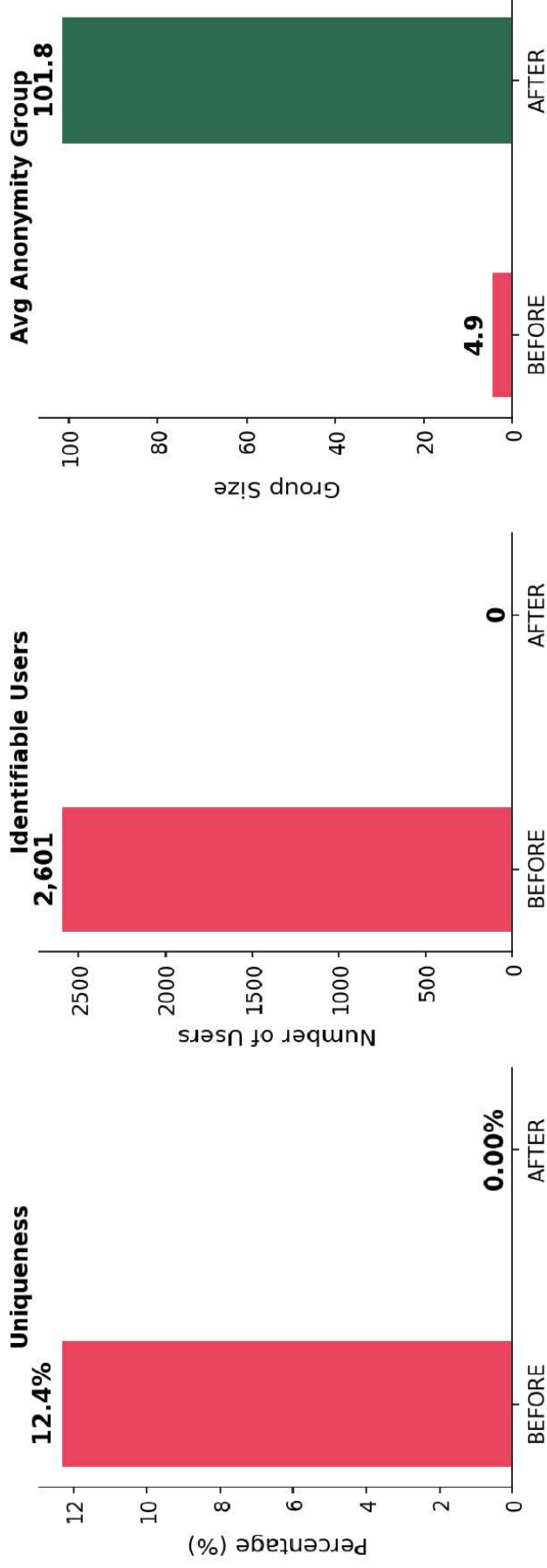
- Differential privacy adds **mathematical noise** to query results
- Benefits:
 - Protects individual identities
 - Preserves overall statistical patterns
 - Prevents reconstruction attacks
- Synthetic data generation can also produce **realistic but non-identifiable** datasets.



Privacy improvement

Impact of Mitigation:

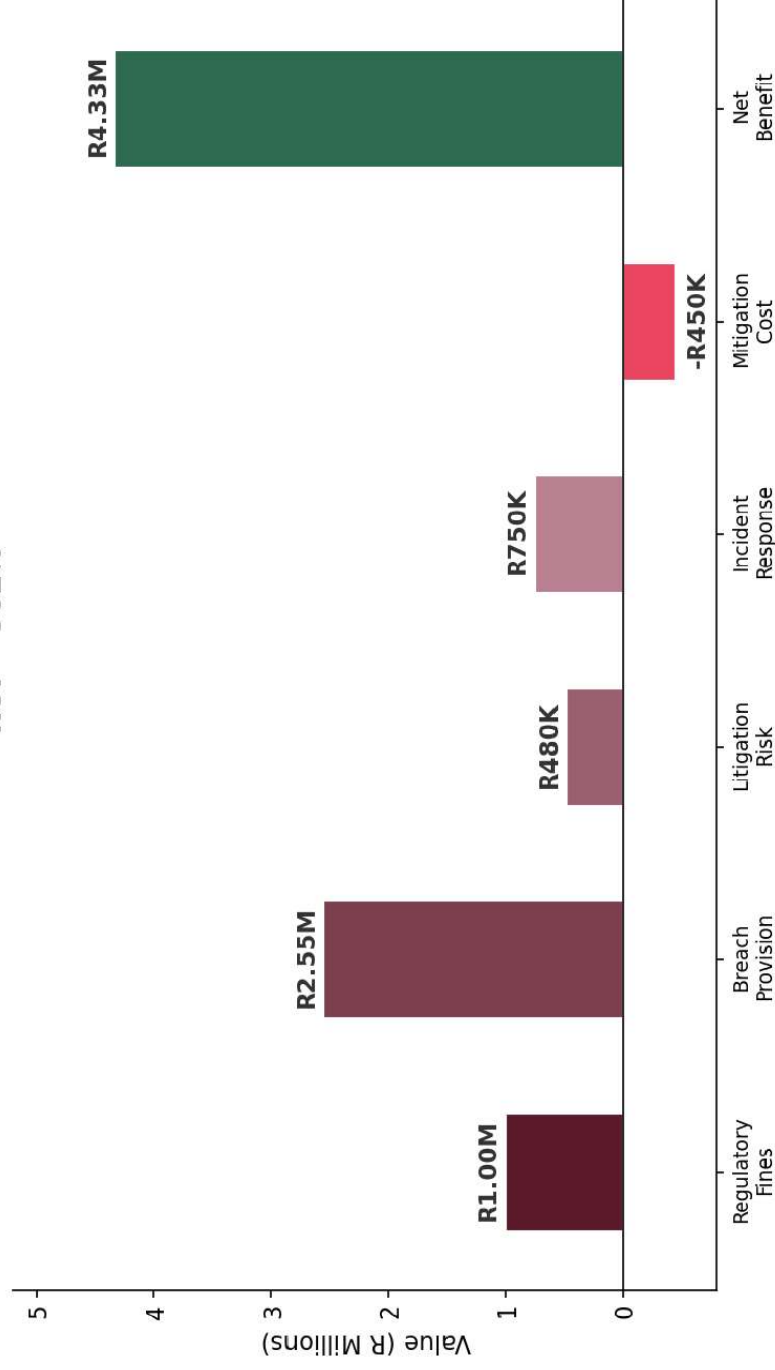
Privacy Improvement: Before vs After Mitigation



Privacy risk dramatically reduced while maintaining analytical value.

Risk-Adjusted ROI: Privacy Controls vs Expected Breach Costs

Risk-Adjusted ROI: Privacy Controls vs Expected Breach Costs
ROI = 962%



Risk Component	Calculation	Expected Value (R)
Regulatory fines	10% x R10,000,000	R1,000,000
Data breach provision	15% x R17,000,000	R2,550,000
Litigation exposure	8% x R6,000,000	R480,000
Incident response & remediation	Estimated	R750,000
ROI (Net benefit/cost)	R4330000/R450000	962%

References (including Task A)

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